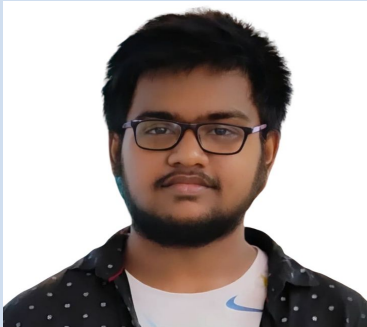


IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – **Startup Blueprint Generator Agent**

Domain of Project – **Agentic AI(Business Strategy And Planning)**

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Problem Statement No.20 – Startup Blueprint Generator Agent

Problem Statement

Entrepreneurs often struggle with the ambiguity of early-stage planning. The Startup Blueprint Generator Agent addresses this by utilizing RAG (Retrieval-Augmented Generation) to convert raw concepts into precise, actionable business roadmaps. By synthesizing market research, competitor data, regulatory landscapes, and funding avenues, the agent provides comprehensive guidance—ranging from Business Model Canvases and financial forecasting to go-to-market strategies—significantly accelerating the startup launch lifecycle.

Project Objectives

The goal is to engineer an AI-driven agent capable of producing data-backed, strategic business frameworks. Core functionalities include:

- **Holistic Business Architecture:** Automated generation of detailed 19-section business plans, featuring executive summaries and SWOT analysis.
- **Contextual Market Intelligence:** Intelligent summarization of industry-specific compliance requirements, government schemes, and localized funding opportunities.
- **Dynamic Financial Modeling:** Tailored budget projections and unit economics adjusted to the specific scale of the venture.
- **Investor-Ready Tooling:** Automated drafting of professional pitch decks and Business Model Canvases (BMC).
- **Documentation & Visualization:** A responsive dashboard for real-time progress tracking and data visualization, alongside one-click professional PDF report generation.

Proposed Solution - AI-Powered Startup Blueprint Generator Agent

The proposed solution is an autonomous, AI-driven Startup Blueprint Generator designed to dismantle the significant knowledge and financial barriers hindering entrepreneurship. By leveraging IBM watsonx.ai and IBM Granite models, the platform provides comprehensive, real-time strategic guidance for new ventures.

Core Methodology

The system architecture utilizes a Retrieval-Augmented Generation (RAG) framework to ensure strategic outputs are grounded in verified, real-world data. By pulling accurate insights from authoritative economic and institutional sources—including current market trends, government schemes, and legal regulations—the Granite foundation models deliver contextually precise and reliable business strategies.

Operational Logic: The Four Virtual Agents

The system logic is divided into four specialized agents:

- **Market Research Agent:** Identifies target audience demographics, performs deep competitor analysis, and tracks industry-specific market trends.
- **Financial Planning Agent:** Develops estimated budget breakdowns, pricing strategies, unit economics, and scalable revenue models customized to the startup's scale.

- **Compliance & Legal Agent:** Maps out critical legal frameworks, mandatory business registrations, and region-specific government incentive programs (e.g., Startup India or SBA).
- **Go-To-Market & Pitch Agent:** Generates actionable 30-60-90 day execution roadmaps, Business Model Canvases (BMC), and professional investor pitch templates.

Technical Architecture

The platform features a modern, responsive web dashboard—built with Flask, Bootstrap, and JavaScript—which facilitates secure user input and displays the detailed 19-section blueprint. Backend workflows are managed through Flask and Gunicorn, while the watsonx.ai ecosystem ensures enterprise-grade deployment, scalability, and robust AI governance. A server-side PDF export engine allows for seamless, one-click document sharing.

Conclusion

Ultimately, this platform functions as an end-to-end intelligent incubation assistant, enabling aspiring founders to convert concepts into investor-ready, legally compliant, and financially robust business strategies.

Technology Used

- **Flask & Gunicorn (Python)** – For designing, routing, and orchestrating the complex generation workflows and securely handling backend API requests.
- **IBM Granite model (ibm/granite-3-8b-instruct)** – For deep natural language understanding, business logic reasoning, financial planning, and personalized text generation.
- **IBM Cloud** – Provides the underlying infrastructure to build, deploy, and scale the Startup Agent efficiently. IBM Cloud enables a reliable, secure, and scalable environment for enterprise deployment.
- **RAG (Retrieval-Augmented Generation) Framework** – To dynamically fetch required business structures, country-specific government schemes, and market definitions before generating intelligent answers, making the Startup Blueprint highly trustworthy and actionable.
- **IBM Watsonx.ai** – For enterprise-grade model access, API deployment, and robust AI governance.
- **FPDF2 & Bootstrap 5** – For dynamic, server-side PDF document rendering and building a highly responsive, interactive frontend UI visualization.

Architecture Components Used -

- 1) **User Input Interface** – A modern, responsive web form where users enter their raw startup idea, target industry, operating country, and initial budget.
- 2) **IBM watsonx AI Agent** – Processes the founder's input using IBM Granite models to autonomously research and generate a comprehensive 19-section strategic business plan.
- 3) **Name of IBM Granite model** – `ibm/granite-3-8b-instruct` → Highly efficient, enterprise-grade model optimized for complex business logic, financial reasoning, and strategic planning.
- 4) **Interactive Output Dashboard** – Elegantly displays the AI-generated responses (such as the Business Model Canvas, SWOT analysis, and Go-to-Market strategy) in a structured, tabbed web interface.
- 5) **PDF Export Component (FPDF2)** – Instantly compiles and formats the generated text into a downloadable, professional A4 PDF document ready for investor pitches.
- 6) **Python Backend Router (Flask)** – Manages user sessions and dynamically splits massive data requests to bypass AI token limits, ensuring seamless, high-speed document generation.

Role of Agentic AI in the Startup Blueprint Generator

Agentic AI enables the Startup Blueprint Generator to function as an intelligent, autonomous business strategist rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates a complex, multi-step reasoning process to deliver a comprehensive 19-section business plan.

The AI performs highly specialized roles—such as retrieving target market demographics, formulating competitive analyses, structuring financial revenue models, and identifying country-specific legal compliance. By breaking down a massive strategic task into specialized cognitive steps, this agentic approach ensures efficient execution and highly accurate results.

Agentic AI also brings deep **context awareness**, deeply understanding specific user constraints like their initial budget, operating country, and target industry to generate hyper-tailored recommendations. It bypasses standard generic advice and applies real-world economic logic.

Additionally, the system formats and structures the output dynamically, ensuring that the generated financial models align perfectly with the proposed go-to-market strategies for cohesive decision-making.

Overall, Agentic AI makes the system proactive, analytical, and goal-driven, acting like a **smart digital startup incubator** that helps aspiring founders launch successfully.



Novelty and Uniqueness

The proposed Startup Blueprint Generator stands out by combining Agentic AI, dynamic financial modeling, and automated document formatting into a single intelligent incubation system.

- **Autonomous Strategic Reasoning** – Instead of a simple Q&A chatbot, the system autonomously reasons through 19 complex business dimensions (from SWOT to Go-To-Market) to deliver a cohesive, end-to-end business strategy.
- **RAG-Based Accuracy** – Integrates targeted retrieval of country-specific legal frameworks and government incentive schemes, significantly reducing AI hallucinations and ensuring compliant, realistic business plans.
- **Hyper-Personalization** – Tailors the entire business model, pricing strategy, and target audience based entirely on the founder's specific industry, operating country, and initial budget.
- **Instant Professional Export** – Features a built-in server-side PDF rendering engine, allowing founders to instantly download formatted, investor-ready documents with a single click.
- **Dynamic Web Dashboard** – Provides a seamless, interactive UI with animated generation tracking and tabbed reading, offering a vastly superior user experience compared to standard terminal-based AI tools.
- **Financial Literacy Focus** – Goes beyond basic startup advice to provide actual unit economics, estimated budget allocations, and scalable revenue models for first-time founders.
- **Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring enterprise-level reliability, data privacy, and governance.

GitHub Link

Github URL:

https://github.com/RajKiranAcharyya/AICTE_IBM_EDUNET_INTERNSHIP_Startup_Blueprint_Generator

Live Web App URL: <https://startupplan-generator-edunet-foundation.onrender.com>

Demo Video:

https://docs.google.com/videos/d/1r0zR4eqxMDFpNzph_9MeQwZTrGSSQXVg3asL4nH2F_A/play

Project Screenshots

Startup Blueprint Generator — | x +

Ask Gemini | — □ ×

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 **StartupBlueprint.AI**

Powered by IBM watsonx.ai

🌙

startup plan.

 **Startup Idea ***

A subscription-based eco-friendly stationery brand that delivers high-quality notebooks, planners, pens, and office supplies made from recycled and sustainable materials. Customers can customize products, receive monthly stationery boxes, and purchase through an online store.

276 / 500

 **Business Goal ***

Reach 15,000 customers within two years, establish partnerships with schools and businesses, achieve profitability by the end of Year 2, and expand across India.

161 / 300

 **Industry ***

E-Commerce

 **Country ***

India

 **Target Audience ***

Students, professionals, schools, colle

 **Available Budget ***

\$5,000 – \$25,000

 **Generate My Blueprint**

 Generation typically takes 30–60 seconds. Please wait while IBM Granite analyses your idea.

Project Screenshots



✓ Blueprint Generated Successfully!

A cloud kitchen that prepares fresh, healthy, and affordable meals for...

FoodTech · India · Generated July 12, 2026 at 08:29 PM

 View Full Report

 Copy Report

 New Blueprint



Section 1 Executive Summary

The proposed startup is a cloud kitchen that prepares fresh, healthy, and affordable meals for working professionals in India. With an initial investment of \$5,000 to \$25,000, the goal is to serve 1,000 daily orders with...



Section 2 Business Model Canvas

The business model canvas for the cloud kitchen startup consists of the following key components: Customer Segments Working professionals aged 22-45 Fitness enthusiasts Busy families Health-conscious individuals in urban...



Section 3 Market Research

The Indian FoodTech industry is growing rapidly, with an estimated market size of \$15 billion by 2025. The demand for healthy and convenient meal solutions is increasing, driven by the rising awareness of health and well...



Section 4 Competitor Analysis

The Indian FoodTech industry is highly competitive, with several established players and new entrants. Some of the key competitors in the cloud kitchen space include: Key



Section 5 SWOT Analysis

The cloud kitchen startup has several strengths, weaknesses, opportunities, and threats that will impact its success in the Indian FoodTech industry. Strengths Unique



Section 6 Target Customers

The target customers for the cloud kitchen startup are working professionals aged 22-45, fitness enthusiasts, busy families, and health-conscious individuals in urban

Project Screenshots

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- 11 Marketing Strategy
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- 13 Funding Opportunities
- 14 Government Startup Schemes
- 15 Legal Checklist
- 16 Risk Analysis
- 17 Growth Roadmap
- 18 Investor Pitch
- 19 30-60-90 Day Action Plan

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SECTION 1

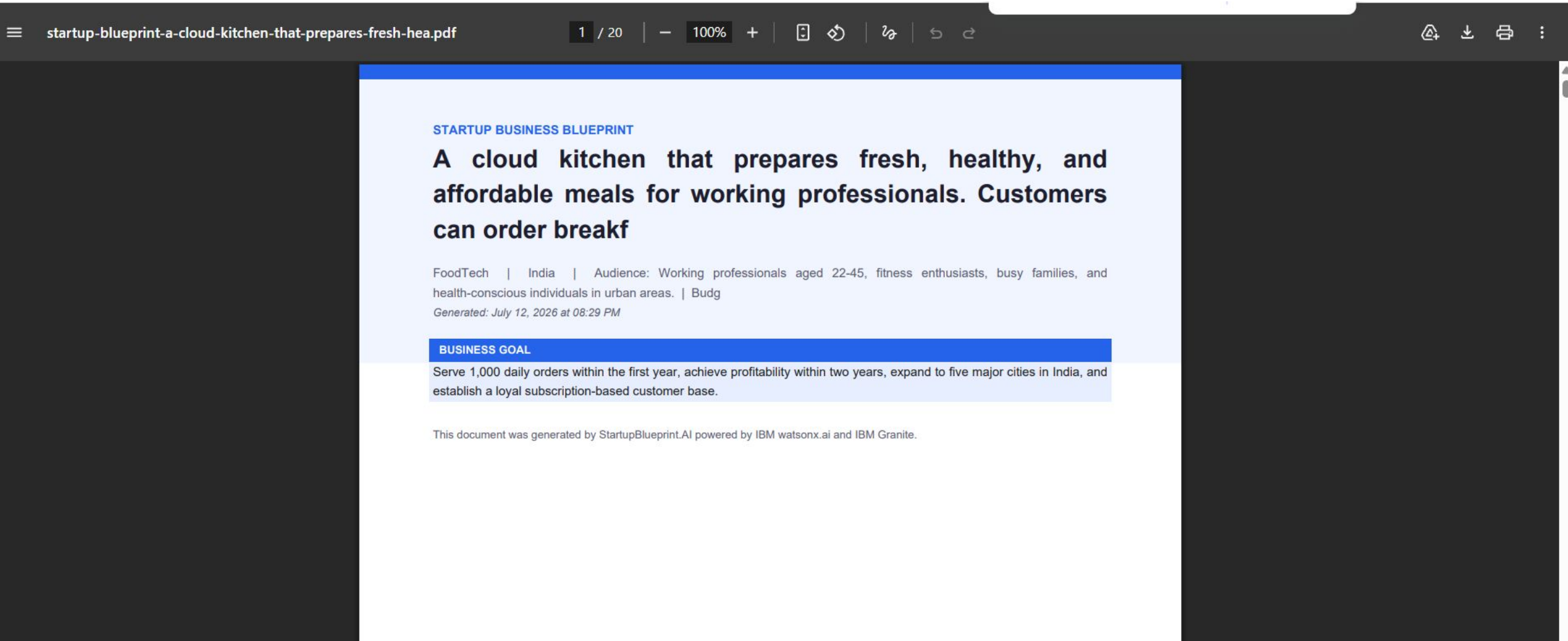
Executive Summary

The proposed startup is a cloud kitchen that prepares fresh, healthy, and affordable meals for working professionals in India. With an initial investment of \$5,000 to \$25,000, the goal is to serve 1,000 daily orders within the first year, achieve profitability within two years, and expand to five major cities in India. The cloud kitchen will operate through a mobile app, offering breakfast, lunch, dinner, and weekly meal subscriptions with customizable meal plans. The menu will cater to various dietary preferences, including low-calorie, high-protein, vegetarian, and vegan options. By leveraging the growing demand for healthy food and convenient meal solutions, the startup aims to establish a loyal subscription-based customer base and become a leading player in the Indian FoodTech industry.

Key Highlights

- Target market: Working professionals aged 22-45, fitness enthusiasts, busy families, and health-conscious individuals in urban areas
- Initial investment: \$5,000 to \$25,000
- Projected daily orders: 1,000 within the first year
- Projected profitability: Within two years
- Expansion plans: Five major cities in India
- Menu offerings: Low-calorie, high-protein, vegetarian, and vegan options with customizable meal plans

Project Screenshots



Future Scope

1. Integration with Real-Time Financial APIs & Live Market Data – The system can be integrated with live financial APIs (like Bloomberg or Yahoo Finance) and government business registry databases to fetch real-time economic conditions, live competitor pricing, and active grant availability. This will enable the Startup Agent to provide highly dynamic, data-backed financial forecasts and live market trends rather than static estimates.

2. AI-Powered Pitch Deck Generator & Multilingual Support – Enhancing the system to automatically generate full PowerPoint (.pptx) investor pitch decks based on the blueprint, along with support for multiple languages. This will allow founders to seamlessly pitch to international investors in their native languages, making the solution globally inclusive and vastly accelerating the fundraising process for non-English speakers.

Certificates Earned:

Getting Started with Artificial Intelligence credly certificate



Certificates Earned:

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Thank You!

Thank you for your time and interest.